



UNDERSTANDING THE PROBLEM

Triage decides who waits, and in most emergency departments it is executed once, at the door. **Acuity is then treated as fixed while the patient's physiology keeps moving.** Three structural failures follow: judgment varies between nurses, half the room sits inside one crowded colour band with no ordering, and nothing re-measures a patient who was judged safe. A single adult-calibrated vital-sign scale then adds a fourth, because the same numbers mean opposite things at age 3 and age 78.

HOW IT FAILS — ONE PATIENT, NINETY-TWO MINUTES

T = 0

Arrives, triaged once

HR 92 • RR 18 • BP 128/80 • SpO₂ 97%. No chest pain — all four inside the adult band.

ESI-4 • POSITION 11 OF 14

T + 45

Still in the waiting room

No re-measurement is scheduled. No mechanism exists to look at her again.

UNOBSERVED

T + 75

Deteriorating, unseen

Diabetes blunts the cardiac pain that would have raised the alarm.

UNOBSERVED

T + 92

Critical event

Severe multi-vessel coronary disease. Ejection fraction 25–30%.

ESI-1 • TOO LATE

One decision. Ninety-two minutes. Nothing in the department was designed to revisit it.

THREE SEPARABLE GAPS — AND WHAT EACH ONE COSTS

GAP 01 • VARIABILITY

0.46–0.91

inter-rater agreement

Two competent nurses assign different levels to the same patient. Judgment, not protocol, decides.

OUR MECHANISM Deterministic acuity floor: four rule layers combined with min().

GAP 02 • QUEUE BLINDSPOT

50%+

of visits in one band

Inside a colour, arrival time decides position. A crowded ESI-3 band has no ordering at all.

OUR MECHANISM Continuous within-colour ranking by risk and trend.

GAP 03 • STATIC MEASUREMENT

92 min

unobserved, case P01

Triage executes once. Deterioration in the waiting room is invisible until someone walks past.

OUR MECHANISM Re-triage loop on a timer, plus two mandatory triggers.

WHY NOW — THE EVIDENCE

66%

sensitivity of human triage for patients who go on to need a life-saving intervention. One in three is not identified.

Sax et al., JAMA Netw Open 2023 • 5.3 min encounters, 21 hospitals

50%

of arrivals have no prior record on file, so history cannot be assumed as an input.

Round 2 brief reference parameter

ESI v5

extended vital-sign checks to low-acuity patients explicitly to prevent under-triage.

ESI Handbook v5 • AHA three-tier, prospectively validated

425,000

de-identified ED stays with linked outcomes are now public, so training no longer needs a hospital partnership.

MIMIC-IV-ED, PhysioNet

Jan 2026

FDA CDS guidance: the clinician must be able to independently review the basis of a recommendation.

FDA CDS final guidance • DPDP Act 2023 names the processor boundary

FRAMING DECISIONS, TAKEN UP FRONT

Jurisdiction

India • DPDP Act 2023

Acuity scale

ESI, five levels

Display

RED / YELLOW / GREEN

Deployment

100–500+ visits per day

Evidence

500 synthetic encounters

WHAT WE ASSUME — AND LABEL AS OURS

Synthetic evidence. All engine figures are measured on data we generated against ground truth we assigned.

Crowding share, 50%. The only invented input in the top-down sizing chain.

Two-thirds haircut. Applied to our own simulated impact before we quote it.



ONE ADULT BAND, TWO OPPOSITE FAILURES

3-YEAR-OLD, FEBRILE

👶 HR 140 · RR 28 · 38.5 °C · alert, capillary refill under 2 s

ADULT-CALIBRATED MODEL RETURNS

NEWS2 7 — emergency response

HR 140 is band-normal for ages 1–4; heart rate rises about 10 bpm per 1 °C of fever. Every observed cue is reassuring.

ERROR: OVER-TRIAGE

Engine returns ESI-3 + fever pathway

78-YEAR-OLD, "NORMAL"

👴 HR 95 · RR 18 · 37.0 °C · BP 112/68 · new confusion

ADULT-CALIBRATED MODEL RETURNS

Every parameter scores 0 — GREEN

Blunted fever response. Beta-blockade caps the tachycardia: HR 95 here is the equivalent of 130. Systolic 112 against a baseline of 150 is a 25% drop.

ERROR: UNDER-TRIAGE

Engine returns ESI-2 + sepsis screen

THE MECHANISM WE BUILT INSTEAD

Five vital-sign bands. 0–1, 1–4, 5–11, 12–17 and adult, plus a ≥ 65 profile working on deviation from the patient's own baseline.

Deviation, not raw values. Features are engineered as distance outside the patient's own age band — which is what makes the monotonic constraint clinically meaningful.

Observed cues are first-class. ACVPU, work of breathing, colour and gait are structured inputs. New confusion alone scores +3 on NEWS2; a vitals-only pipeline drops it.

ONE SET OF OBSERVATIONS, FOUR DIFFERENT ANSWERS

SAME NUMBERS,
FIVE AGE BANDS

The calibration thesis

HR 92 · RR 18 · SBP 128 · SpO₂ 97 · 36.9 °C — held constant, scored in each band.

IF THIS PATIENT WERE

LEVEL

ENS

Infant · 6 mo

■ ESI-2

5

Child · 3 y

■ ESI-3

4

Child · 8 y

■ ESI-4

1

Adult · 34 y

■ ESI-5

0

Geriatric · 78 y

■ ESI-2

0

One set of observations, 4 different answers. A single adult-calibrated scale would return one, computed by the 35 mirror — hypothetical ages, no measured record.

Live from the prototype: the same observations held constant and scored in each age band. A single adult-calibrated scale would return one answer for all five.

WHY THE OLD PATTERN PERSISTS

Diabetes blunts cardiac pain. Up to 40% of infarctions in diabetics present atypically — the alarm signal is absent, not weak.

Older women present atypically. Nearly half of over-65s with an infarction arrive with no chest pain at all.

The benign pattern is more available. GI symptoms read as gastritis under time pressure, because gastritis is far more common.

WHAT THE CONTEXT LAYER IS WORTH

HIGH-ACUITY PATIENTS MISSED · SAME 500-ENCOUNTER COHORT

Adult-calibrated vitals only

45.0%

Risk modifiers removed

20.0%

Full engine as proposed

7.5%

Six times fewer misses at the same over-triage cost.

Over-triage held at 33.2% in the vitals-only baseline and 32.0% in the full engine — the improvement is not bought with extra false alarms. Removing the context layer nearly triples misses, so the layer is not a refinement: it is the product.

MEASURED, NOT ASSERTED

16 of 21

hidden-sick caught
genuinely high-acuity on
entirely normal vitals

0

monotonicity violations
8 features × 1,200 held-out
rows

0.943

held-out AUROC
AUPRC 0.838 · Brier 0.056 ·
logistic baseline 0.934

3.0%

encounters abstain
confidence below 0.50 routes to
a clinician

SYNTHETIC · NOT CLINICAL

Ground truth is constructed at generation, independent of the engine. These figures show the architecture behaves as designed — not that it helps patients. That claim can only be earned in Phase 0.

FIVE USERS, ONE PRODUCT — AND FIVE DIFFERENT DEFINITIONS OF SUCCESS

TRIAGE NURSE	CHARGE NURSE	ED PHYSICIAN	QUALITY / NABH	HOSPITAL CIO
OWNS The first three minutes.	OWNS The waiting room.	OWNS The authority to move a patient down.	OWNS Accreditation and incident review.	OWNS Integration risk and data liability.
GETS A ranked recommendation, three plain reasons, a confidence figure, and one-click accept or override.	GETS Live queue depth, safe-wait breach countdowns, deterioration escalations, and surge alerts batched into one channel.	GETS The reasoning trace, the confidence figure, and an explicit prompt wherever data is unresolved.	GETS An append-only, hash-chained audit trail plus under-triage reporting broken out by subgroup.	GETS A deployment tier matched to actual maturity: on-premise, with no PHI leaving the hospital network.
SUCCESS Time to decision falls — and the tool is actually used, not worked around.	SUCCESS Zero un-actioned safe-wait breaches.	SUCCESS Override rate sits in a healthy band — in both directions.	SUCCESS Tamper-evident evidence exists for every decision.	SUCCESS Deployed without an integration project.

WHO THIS IS EXPLICITLY NOT FOR — AND WHY THE BOUNDARY IS DELIBERATE

NOT THE PATIENT No self-triage. It shifts liability onto the person least equipped to carry it.	NOT A DIAGNOSTIC ENGINE It recommends a test and a priority. It never names a diagnosis.	NOT ADMIN-FIRST Throughput improves as a consequence of safety. Optimising only wait time teaches the model to down-rank.	NOT A STAFFING SUBSTITUTE It changes queue order, not queue length. Capacity is a separate problem.
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THE ALTERNATIVES WE REJECTED — AND WHAT WE TOOK FROM EACH ONE ANYWAY

FROM "DO NOTHING" Beat the human baseline before touching a patient. That requirement became Phase 0 — a retrospective gate on records that already exist, with no clinician seeing an output.	FROM "HIRE TWO MORE TRIAGE NURSES" Staffing and this tool are complements, not substitutes. So we measure misses avoided — never nurses replaced.	FROM "BUY A COMMERCIAL AI TRIAGE PRODUCT" Every architectural constraint is tied to a published failure mode. The acuity floor, the monotonicity guarantee and the abstention lane each answer a documented failure of a shipped system.
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WHAT ADOPTION ACTUALLY REQUIRES — THREE CONDITIONS, ALL OF WHICH MUST HOLD

THE NURSE HAS TO TRUST IT A ranked recommendation, three plain reasons and one tap. Measured on time to decision and on actual use — a tool that is worked around has failed, whatever it scores.	THE PHYSICIAN HAS TO BE ABLE TO OVERRULE IT Override is one click and no form, and costs exactly what accepting costs. The rate is monitored in both directions: below 2% or above 30% triggers review.	THE HOSPITAL HAS TO BE ABLE TO AUDIT IT An append-only, hash-chained trail with under-triage reporting broken out by subgroup — evidence that survives an incident review, not a dashboard.
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The product only works if the nurse trusts it, the physician can overrule it, and the hospital can audit it.



THE DETERMINISTIC ACUITY FLOOR



MACHINE LEARNING

Monotonic GBM finds hidden risk the rules cannot see.



MAY RAISE ACUITY

ESI-4 → ESI-2, without asking permission

DETERMINISTIC ACUITY FLOOR

ESI Point A - red-flag rules - age & history modifiers - NEWS2 five bands

min ()



NO PATH DOWN EXISTS IN CODE

No component can lower a level



LICENSED CLINICIAN — THE ONLY DOWNWARD PATH

Reason code plus re-measured evidence, under their licence number.

THE INVARIANT - MAY RAISE, NEVER LOWER

The system is allowed to be cautious. It is not allowed to be falsely reassuring — and that is a property of the code, not of the training run.

WHEN THE ENGINE DOES NOT KNOW

Missing data is not permission to guess. Below 0.50 confidence the engine abstains, holds the level rather than lowering it, and routes the case to a clinician — naming the test that would resolve the uncertainty, never a diagnosis.

0.47

confidence, case P14

2.8%

missed - with a record

11.4%

missed - no record

WHY THE GUARANTEE HOLDS



Enforced by construction

The four rule layers combine with min(); the learned score is clamped to the result. It is a constraint, not a weighting we hope holds.



Provable without a test set

Monotonicity is a property of the code, so the guarantee holds on data the model has never seen — including the day the case reix shifts.



Verified - zero violations

Every clinical feature raised by 1 SD across 8 features = 1,200 held-out rows. Worse physiology never once scored safer.

TRAINED ON OUTCOMES, NOT ON OPINIONS

Train on the acuity a human assigned and you teach the model to reproduce the miss. PO1 was assigned low acuity by a competent nurse.

Labels come from what happened next — ICU transfer, admission, death, or intervention within 24 hours.

WHAT WE WILL NOT CLAIM

- Synthetic training proves the pipeline, not the medicine.
- Real data will need recalibration — Boston 2011–2019 in Uttar Pradesh is distribution shift.
- Subgroups are audited before deployment; 0.941 paediatric, 0.945 adult, 0.935 geriatric.
- No claim of clinical efficacy is made anywhere in this deck.

WHAT THE ARCHITECTURE INHERITS

Acuity scale	ESI v5, four decision points
Local protocol	AIIMS three-tier, validated
Vital scoring	NEWS2, five age bands
Training data	MIMIC-IV-ED, ~425,000 stays
Regulatory test	FDA CDS guidance, Jan 2026
Data law	DPDP Act 2023 - ABDM policy

SOLVED — AND NOT SOLVED

Acuity scoring	solved
Safety architecture	not solved
Deployability	not solved
Clinical trust	not solved

THREE CHEAPER ANSWERS, AND WHY THEY FAIL

Do nothing. One in three critically ill patients is not identified, and the failure is invisible to the department it happens in.

Hire two more triage nurses. Buys throughput, not consistency — two more sets of judgment ranging 0.46–0.91, and none of them re-checks the waiting room.

Buy a commercial AI triage product. The canonical deployed example scored AUC 0.63 externally, missed 67% of sepsis, and alerted on 18% of all patients.

Every architectural constraint in this deck derives from a published failure of a shipped system.



PROBLEM

INSIGHT

SOLUTION

PRODUCT

ARCHITECTURE

IMPACT

EXECUTION

PatientTriage.ai

THE WAITING ROOM — RANKED, AND RE-SCORED ON A TIMER

QUEUE · RANKED

☐ FIRST-COME ORDER 1 min / 2 s Pause

#	PATIENT	ACUITY	WAITED	RE-SCORE	RISK
1	Shanti K. P19 · 63y F · Bukhar and weakness since two days.	RED ESI-2	40m BREACH	14m 2x DONE	58% RISK
2	Kamla Devi P05 · 82y F · Not herself since yesterday, not eating. Beti says she	RED ESI-2	58m BREACH	13m 1x DONE	45% RISK
3	Nafisa S. P21 · 73y F · Chakkar and kamzori, could not get up this morning.	RED ESI-2	27m BREACH	13m 1x DONE	48% RISK
4	Deepak M. P12 · 52y M · Bukhar and cough four days. On chemotherapy.	RED ESI-2	29m BREACH	13m 1x DONE	40% RISK
5	Ramesh K. P02 · 47y M · Seene mein dard aur pasina, one hour.	RED ESI-2	16m BREACH	13m 1x DONE	37% RISK
6	Gopal S. P15 · 71y M · Weakness and dizziness. Dialysis on Tuesday.	RED ESI-2	43m BREACH	13m 1x DONE	23% RISK
7	Meena R. P09 · 26y F · Bleeding and pet dard, six weeks pregnant.	RED ESI-2	11m BREACH	13m 1x DONE	18% RISK
8	Fatima B. P11 · 68y F · Fell in bathroom, hit head. On blood thinners.	RED ESI-2	21m BREACH	13m 1x DONE	14% RISK
9	Sunita D.	RED ESI-2	94m	13m	12%

① Rank + movement

The ▲ badge shows how far a patient moved since the last re-score.

② Acuity, never colour alone

RED / YELLOW / GREEN always ships with the ESI level as text.

③ Safe-wait breach

Waited time turns red when it passes the safe window for that level.

④ Re-score counter

Every patient carries how many times the timer has re-evaluated them.

Shanti K. arrived stable and moved thirteen places to #1 on a re-score — no nurse had to walk past her.

THE CLINICIAN'S JOURNEY, END TO END

1 · INTAKE

Vitals, free text in English or Hinglish, history and observed cues at the desk. Three taps, no typing on shift.

2 · SCORE

Four rule layers run deterministically on-device in under 800 ms. No network call in the clinical path.

3 · EXPLAIN

RED / YELLOW / GREEN with an ESI level, a confidence figure and at most three plain reasons — or an explicit abstention.

4 · DECIDE

Accept with one tap, or override with one tap and a pre-populated reason code. Both cost the same.

5 · WATCH

The whole waiting room is re-scored on a timer and on every new observation; ranking updates itself.

6 · PROVE

Every decision and override writes an append-only, hash-chained record with model and ruleset version.

WHY — THE EXECUTION TRACE BEHIND EVERY SCORE

RECOMMENDATION 8.4 86 · js-mirror, live model v3.1

RED ESI-2 what-if · not a measured record P01 · KDH-2026-118834

confidence 9.72 · Moderals · 8.50-0.79

completeness 1.00 agreement 8.48 margin 0.38

Diabetes ≥10 y · atypical ACS pattern
Cardiac pain blunted — absence of chest pain is not reassurance

ECG within 10 minutes; troponin. Do not wait for chest pain to appear.
The engine recommends a test and a priority. It does not name a diagnosis.

WHY — EXECUTION TRACE min() over layers

DETERMINISTIC · NO NETWORK CALL

A Life-saving intervention
ESI decision point A no match

B Red-flag patterns
rule matcher · 24 named patterns no match

C Risk modifiers
age · comorbidity · presentation context → ESI-2

D Vital-sign scoring
NEMS2 · Adult · 12-64 y no match

ACUITY FLOOR · MIN() · STRICTEST WINS ESI-2

Learned model · monotonic GBM. The model scored risk 12% → ESI-5. It is above the floor, so the floor stands. It can move this arrow up. There is no downgrade path in the code — only a licensed clinician can lower a level, with a reason code.

Language layer read the note as english and discarded 1 negated term (chest pain) — "no chest pain" is not chest pain.

Every recommendation opens its own reasoning: which of the four deterministic layers fired, the acuity floor min() produced, the confidence figure and its band, and the note that the learned model can move the arrow up but has no downgrade path. No score is ever returned without a



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PatientTriage.ai

1 · TRIAGE INTAKE — THE FIRST THREE MINUTES, THREE TAPS, NO TYPING

PatientTriage.ai / TRIAGEOffline · ruleset v1.0

P01 · 58 F · WALK-IN

vomiting · weakness · no chest pain

MEASURED VITALS

92

HR

18

RR

128/80

BP

97%

SpO₂

All four inside the adult band.

RISK MODIFIERS

diabetes · 12y

female

atypical cardiac

"Diabetes + atypical GI presentation increases missed-ACS risk."

CONFIDENCE BAND · MODERATE (0.50–0.79)

Amber chip; contributing factors expanded by default. Below 0.50 the engine abstains and routes to a rapid clinician lane instead.

ACCEPT

OVERRIDE

RECOMMENDATION

RED

ESI-2

confidence 0.72

WHY

risk modifier

context layer

1 · Three taps

Vitals, cues, self-report — no typing on shift.

2 · Colour + level

RED / YELLOW / GREEN always ships with the ESI level as text.

3 · At most three reasons

Plain language, capped, never a diagnosis.

4 · Accept or override

Both cost one tap. Downgrade needs a reason code.

3 · CLINICIAN REVIEW — THE ONLY PATH THAT LEADS DOWNWARD

PatientTriage.ai / PATIENT REVIEWP18 · 55 M

Overview

Why

Vitals trend

Audit

SYSTEM RECOMMENDATION

YELLOW · ESI-3

Arrival HR 122

after climbing stairs + arguing at registration

Downgrade path in code

NONE

Only the clinician can do this.

WRITTEN TO THE RECORD

clinician_id + licence

action · final_acuity

reason_code · evidence_ref

EVIDENCE → OVERRIDE

1 Arrival — HR 122; system recommends ESI-3

2 Re-measure — HR 84 after 5 minutes' rest

3 Reason code — TRANSIENT_PHYSIOLOGY

4 Final — GREEN · ESI-4, clinician-attributed

WHY PANEL

Triggered rules[]

Age band applied

Confidence

Input provenance

Ruleset + model version

CONFIRM OVERRIDE

Reason code + evidence required.

1 · No code path down

No component can lower acuity — only a licensed clinician.

2 · Evidence, not opinion

A downgrade needs re-measured evidence attached.

3 · Attributed

Clinician ID and licence are written to the record.

4 · One click, no form

Override is deliberately as cheap as accepting.

CASE P01 · SAME INPUTS, TWO VERDICTS

ADULT-VITALS BASELINE

LOW PRIORITY

position 11 of 14

92-minute wait

PATIENTTRIAGE.AI

RED · ESI-2

confidence 0.72

caught by risk modifiers

WHICH OF THE FOUR LAYERS FIRED

01 Life-saving checkno match

02 Red-flag patternsno match

03 Risk modifiers→ ESI-2

04 Vital-sign scoringno match

OUTCOME AT 24 HOURS

Severe multi-vessel coronary disease · ejection fraction 25–30%.

WHY FOUR NORMAL NUMBERS ARE NOT REASSURANCE

Diabetes blunts cardiac pain · up to **40% of MIs in diabetics** present atypically · women present atypically · **nearly half of over-65s** arrive without chest pain · GI symptoms read as benign.

7.5% → 20.0%

high-acuity missed, with the context layer removed

ON THE SUBGROUP THAT HIDES

Patients who look well on vitals alone: **5 of 21 missed** with the modifiers in place, **15 of 21 missed** without them.

The output stays a test and a priority. **It never names a diagnosis** — the atypical-presentation evidence changes the queue position, not the label.

UNDER SURGE THE CLOCK MOVES — THE THRESHOLDS DO NOT

RED / YELLOW

re-score 15 → 5 min

GREEN

re-score 30 → 20 min

Alerts

pop-ups batch into one dashboard

Thresholds

never move

P19 — ESI-3 → ESI-2 at T+38 min as EWS goes 2 → 6; escalated by the timer, not by a nurse walking past.

P14 — no record, confidence 0.47: the engine holds ESI-3 and routes to the rapid clinician lane.

EVERYTHING ELSE DEGRADES HONESTLY

At 3× volume peak queue multiplies **22×** and **66%** of safe-wait windows are breached. No threshold, rule or floor moves under surge — only how often we look.

Four layers saw nothing. One caught her — and one is enough.

Illustrative MVP wireframes rebuilt from the working prototype and the deck's example cases. UI labels are presentation scaffolding; clinical facts and patient values are unchanged. Team Claudemax69 · IIT Kanpur · Accenture Innovation Challenge 2026, Round 2, Problem Track 2.

No hospital needs Tier 2 to get the Tier 0 safety benefit. Encounter and Location **Runs on one tablet – no GPU, no cloud, no API latency.**

THE DECISION RECORD — WHAT IS WRITTEN, WHO OWNS IT, AND WHAT IT PROTECTS AGAINST

EVERY DECISION WRITES EXACTLY ONE RECORD

01 · EVENT

event_id · timestamp · encounter_id

model_version · ruleset_version · inputs_hash

02 · RECOMMENDATION

recommended_acuity · confidence

triggered_rules[] · age_band

03 · HUMAN ACTION

clinician_id + licence · action

final_acuity · reason_code · evidence_ref

Append-only · hash-chained · WORM-backed

Every decision is reproducible from the log, at the ruleset and model version that produced it.

DECISION LOG · AS WRITTEN

safe-wait breach

evt_00041

P05

ruleset r2026.08.14

re-triage escalation

evt_00042

P19

ESI-3 → ESI-2

conf 0.68

clinician override

evt_00043

P18

ESI-3 → ESI-4

TRANSIENT_PHYSIOLOGY

abstention

evt_00044

P14

conf 0.47

rapid clinician lane

Illustrative rows in the format the prototype writes.

THE LEGAL BOUNDARY, AND WHO STANDS WHERE

Hospital = Data Fiduciary | Engine = Data Processor

DPDP Act 2023

India

The engine acts only on documented instruction from the hospital.

FDA CDS

Jan 2026

The clinician can independently review the basis of a recommendation — the "why" panel is the execution trace.

Deployment

On-premise. Personal data does not leave the hospital network.

Other markets

HIPAA §164.312 · GDPR Art. 9(2)(h) and Art. 22.

Identity never enters the model. Tokenised at ingest; re-identification needs two custodians.

THE ENGINE IS A PROCESSOR, NOT A DECISION-MAKER

It recommends; a named clinician decides. That distinction is what keeps it out of device classification and inside CDS guidance.

A LICENSED CLINICIAN IS THE ONLY DOWNWARD PATH

No component can lower acuity. A downgrade carries a reason code, re-measured evidence, and the clinician's licence number.

NOTHING CROSSES THE HOSPITAL BOUNDARY

Default-deny egress, air-gap capable. Exports are produced inside the network.

THREE RISKS THE ROADMAP SLIDE DOES NOT SHOW

DEVICE RECLASSIFICATION

It recommends a test and a priority, never a diagnosis; a human confirms every recommendation before it takes effect.

Owner · clinical advisor

DATA BREACH

On-premise deployment; pseudonymised at ingest; every read is audited.

Owner · hospital CIO

CLINICIAN REJECTION

Shadow mode runs first; when it goes live, override is one click and no form.

Owner · charge nurse

UNCHANGED FROM THE ROADMAP SLIDE

Patient harm · automation bias · alert fatigue · model drift · distribution shift — each already carries an architectural control and a named owner.

THE CONTROL IS ARCHITECTURAL WHEREVER IT CAN BE

A risk answered by code is answered once. A risk answered by training has to be answered again every intake.

PHASE 0 · THE ABSOLUTE GATE — AND WHAT IT DOES NOT REQUIRE

SILENT RETROSPECTIVE SCORING

The engine scores de-identified historical records. No clinician sees an output.

INPUT

Records that already exist. No new collection, no new consent surface.

OUTPUT

One comparison against the department's own under-triage rate.

GATE

Under-triage below the human baseline — or the correct response is to stop.

No integration

No capital outlay

No workflow change

No patient exposure

No queue is reordered · no patient's care changes

WHAT THE LOG LETS A BOARD ASK

What was known, what was recommended, who decided otherwise — and on what stated ground. Claims are lost on documentation, not only on care.

WHAT NEVER LEAVES

Nothing identifiable leaves the network and no third party sits in a life-safety path. A charge nurse can switch it off without degrading the existing triage process.

The audit trail is not a compliance add-on. It is the artefact that makes an override safe to make — and safe to defend.



ONE DEPARTMENT, ONE YEAR

109,500

annual visits at 300/day

10,950

high-acuity · 10% share

3,723

missed today · human 34%

821

missed with the system · 7.6%

967

additional high-acuity patients identified per year — a two-thirds haircut on the simulated 2,902, because simulation will not transfer intact.

Top-down cross-check: 1.40 bn → 119 mn ED registrations → 60 mn in crowded EDs (est. 50%, ours) → 6.5 mn under-triaged (11%) → 6.5 lakh time-critical at the conservative 10% floor. Halving our crowding assumption still leaves ~3.2 lakh.

FIVE USERS, ONE MEASURE

Triage nurse	A ranked recommendation, three plain reasons, one tap. Used, not worked around
Charge nurse	Queue depth, breach countdowns, one batched alert channel. Zero un-actioned breaches
ED physician	The reasoning trace and the authority to move a patient down. Override rate in a healthy band
Quality / NABH	Hash-chained audit trail, under-triage reporting by subgroup. Tamper-evident evidence
Hospital CIO	A tier matching actual maturity; no PHI leaves the network. Live without an IT project

Not the patient · not a diagnostic engine · not a staffing substitute — it changes the order of the queue, not its length.

Arithmetic: 300 visits/day × 365; high-acuity share at the conservative 10% end of published ESI 1–2 distributions (9.8–14.6%); human miss rate derived from the 66% sensitivity in Sax et al. 2023. Unit costs (₹25,000 ICU day, ₹5,000 ward day, ₹3,500 per retained encounter) are our own assumptions.

TUNED INSIDE THE TRAUMA STANDARD

4.0%

under-triage
ACS-COT ceiling is 5%

32.0%

over-triage
ACS-COT band is 25–35%

CONFUSION MATRIX · 500 SYNTHETIC ENCOUNTERS

	True 1–2	True 3	True 4–5
Pred 1–2	74	45	10
Pred 3	5	167	72
Pred 4–5	1	14	112

Amber cells are under-triage — the only errors that can kill. 20 of 500. Roughly eight over-triages are accepted to prevent one miss: one over-triage costs an avoidable ECG and about twelve minutes of nurse time; one under-triage costs a missed infarction. There is no exchange rate, which is why we never optimise average accuracy.

SYNTHETIC · NOT CLINICAL

AND IT HOLDS UNDER 3× LOAD

MEDIAN WAIT, HIGH-ACUITY PATIENTS · IDENTICAL ARRIVAL STREAMS

First-come order		200.4
Continuously ranked		8.4

Minutes. Mean wait is unchanged by construction — ranking moves the sickest forward, so someone less sick waits longer. That is the trade, and it is the correct one. Every operational number degrades honestly: peak queue depth multiplies 22×, and 66% of safe-wait windows are breached at 3× volume.

No threshold, rule or floor moves under surge. Only how often we look.

THE HOSPITAL BUSINESS CASE

≈ ₹127 lakh

modelled ICU delay avoidance a year

35% of the 967 detections × 1.5 ICU days × ₹25,000. Sensitivity ₹85 L–₹178 L across ₹18k–₹35k per ICU day.

Flat

over-triage cost

32.0% against a 33.2% adult-calibrated baseline — the return is not funded by extra unnecessary workup.

≈ ₹19 lakh

ward capacity released

0.4 days average LOS reduction = 387 bed-days; a further ₹19 lakh modelled from a 0.5 pp LWBS improvement.

₹35,000

Tier 0 deployment

One tablet plus configuration. No integration project, and no licence in Phase 0 or Phase 1.

MODELLED · NOT MEASURED

Modelled payback under 12 months at Tier 2.

AND THE ARGUMENT THAT SURVIVES A BOARD

Liability shield. Claims are lost on documentation, not only on care. Every decision writes what was known, what was recommended, who decided otherwise, and on what stated ground.

No new breach surface. Nothing identifiable leaves the network; no third party sits in a life-safety path.

Reversible. A charge nurse can switch it off without degrading the existing triage process.

NABH-ready. Append-only under-triage reporting by subgroup, with no extraction project.

The return is bought with detection, not with more false alarms.



PROBLEM

INSIGHT

SOLUTION

PRODUCT

ARCHITECTURE

IMPACT

EXECUTION

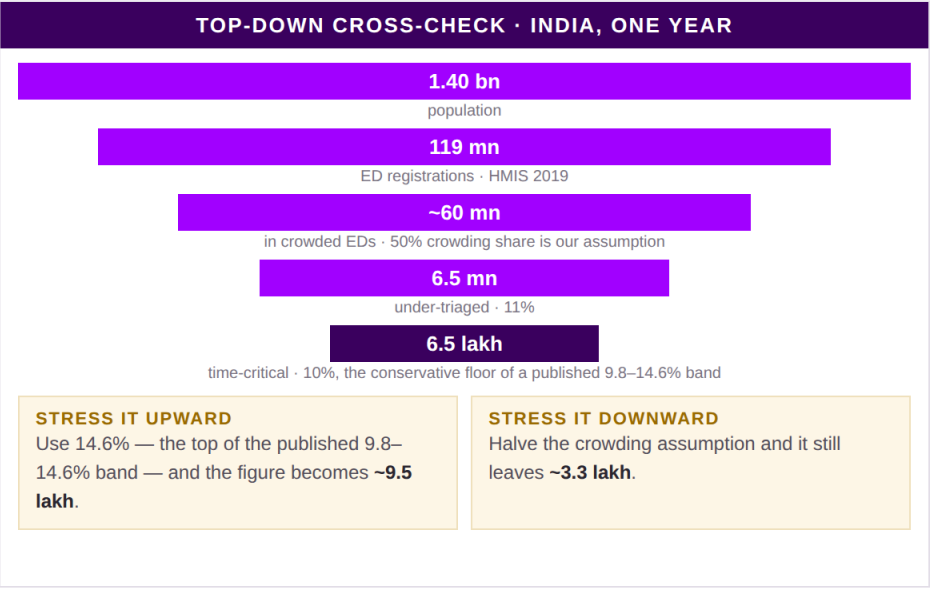
PatientTriage.ai

EVERY NUMBER IN THIS DECK, AND THE ASSUMPTION SITTING UNDERNEATH IT

ABLATION — WHAT EACH LAYER IS ACTUALLY WORTH			
CONFIGURATION	MISSED	OVER	READOUT
Full engine	7.5% (6/80)	32.0%	proposed configuration
Risk modifiers removed	20.0%	—	misses nearly triple
Adult vitals only	45.0%	33.2%	same over-triage cost, ~6× more misses
Hidden-sick subgroup	5/21	—	without modifiers: 15/21 missed
The over-triage column is the point: the context layer removes misses without buying them back as false alarms — 33.2% → 32.0% while misses fall from 45.0% to 7.5%.			
HIGH-ACUITY PATIENTS MISSED · SAME 500-ENCOUNTER COHORT			
Adult vitals only			45.0%
Risk modifiers removed			20.0%
Full engine, as proposed			7.5%

MODEL QUALITY, HELD OUT		
0.943 AUROC	0.838 AUPRC	0.777 sens @ 90% spec
0.058 Brier	0.934 logistic baseline	0 monotonic violations
Cohort 4,000 synthetic encounters · 70/30 split · 2,800 train / 1,200 held out		
Prevalence 16.4%		
Subgroup AUROC 0.941 paediatric · 0.945 adult · 0.935 geriatric		
Monotonicity 0 violations across 8 × 1,200 perturbations		
SYNTHETIC · NOT CLINICAL		

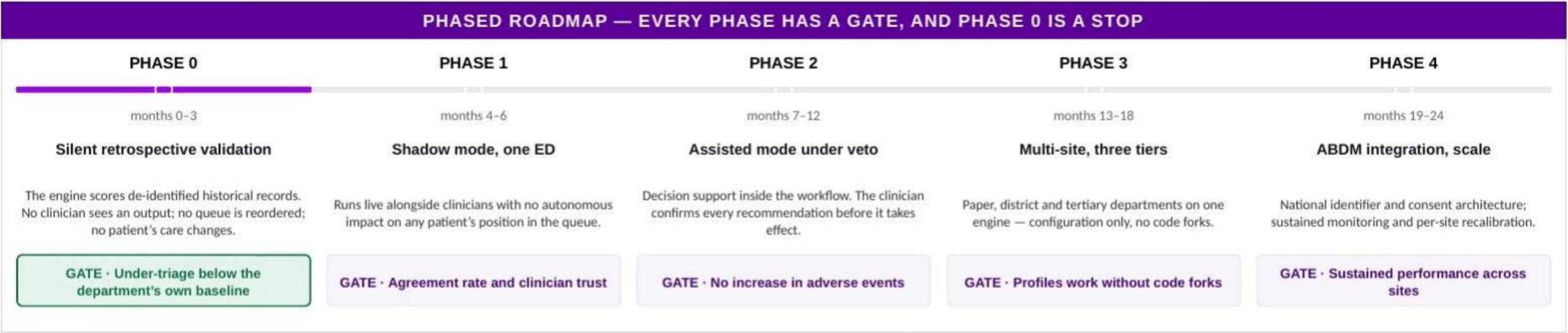
SURGE · 3× ARRIVAL VOLUME, IDENTICAL STREAMS		
MEASURE	NORMAL	SURGE
Arrivals	50	150
Peak queue	4	90
Median high-acuity wait	1.0m	8.4m
95th percentile	3.2m	42.9m
Breach rate	0.0%	66.0%
Mean wait, all patients	4.2m	180.7m
Against first-come at the same surge: median high-acuity 8.4 vs 200.4 min, p95 42.9 vs 326.9 min, breach 66% vs 78%. Mean wait is unchanged by construction — ranking moves the sickest forward, so someone less sick waits longer. That is the trade, and it is the correct one.		



THE SIMULATION, STATED PLAINLY	
Model	Discrete-event · 3 treatment stations · 12 min clinician time · 4-hour window
Load	12.5 arrivals/hour normal vs 37.5 at surge
Over-triage cost	≈ ₹450 and ~12 minutes of nurse time, per event
Exchange rate	~8 over-triages accepted to prevent one miss
There is no exchange rate for one missed infarction, sepsis or child — which is why we never optimise average accuracy.	
REPRODUCIBLE FROM FOUR INPUTS	
01 Visits per day	300 — one mid-size department
02 High-acuity share	10% — the conservative end of published ESI 1–2 distributions
03 Human miss rate	34% — derived from the 66% sensitivity in Sax et al. 2023
04 Engine miss rate	7.5% — measured on our synthetic cohort
Change any one of the four and the whole chain re-derives. Nothing downstream is a fitted constant.	

HOW OFTEN THE ENGINE SAYS "I DON'T KNOW"	
40.6% high ≥ 0.80	55.6% moderate 0.50–0.79
3.8% low < 0.50	
Abstention rate	3.0% of encounters route to a rapid clinician lane instead of a score
Record availability	50/50 in the synthetic cohort
With a record	2.8% high-acuity missed
Without a record	11.4% high-acuity missed

Stress-tested in both directions before we quote it: 14.6% gives ~9.5 lakh, halved crowding gives ~3.3 lakh, and we still plan against the conservative floor.



RISK AND MITIGATION		
Risk	Control — architectural wherever possible	Owner
Patient harm from a wrong recommendation	No component can lower acuity; Phase 0 retrospective gate before any live use	Clinical advisor
Automation bias — the nurse stops thinking	Override rate monitored both ways; below 2% or above 30% triggers review	Charge nurse
Alert fatigue under surge	Alerts batch into one board; three reasons per card, capped; ≤ 6 alerts/nurse/hour	Charge nurse
Model drift after deployment	Monthly recalibration; versioned thresholds; every decision reproducible from the log	Quality lead
Distribution shift across sites	Rules floor is threshold-based and locally owned; recalibration is a Phase 3 gate	Local protocol

THE STRONGEST ARGUMENT AGAINST US — WHICH WE MAKE OURSELVES

Every figure in this deck comes from data we generated, scored against ground truth we assigned.

We accept it. Simulation shows that an architecture behaves as designed; it cannot show that it helps patients. Which is exactly why the roadmap opens with a department’s own historical records, why nothing changes for a patient until that passes, and why the correct response to a failed Phase 0 is to stop — not to re-tune until it passes.

THE ASK — THREE INPUTS, NO CAPITAL

01 One partner emergency department

Any tier. A department running on paper is as useful to Phase 0 as a tertiary centre, because the engine needs no integration to be scored against.

02 De-identified historical records

Complaint, age, sex, recorded vitals, assigned acuity — and what happened next: admission, ICU transfer, intervention within 24 hours.

03 One clinical advisor

An emergency physician or senior triage nurse to review the rule catalogue and hold local ownership of every hard threshold

No integration
scores exported records
offline

No capital outlay
no licence or hardware

No workflow change
nobody sees an output

No patient exposure
retrospective records
only

We are not asking a hospital to trust AI with its patients.

We are asking for the chance to prove whether it deserves that trust.

967 additional high-acuity patients a year, in one 300-visit department — the number we commit to.